GLOBAL ARBITRAGE SIGNAL ENGINE

Internship Selection Challenge — Aquifer Investments

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1 Strategy Design

This project implements a statistical arbitrage framework targeting pricing dislocations across highly correlated ETF pairs. The system exploits short-lived mean-reversion opportunities using a multi-layer signal pipeline comprising cointegration testing, ML-based confidence scoring, and regime-aware filtering.

1.1 Market Inefficiency

The pricing dislocations arise due to several structural factors:

- **Liquidity Imbalance:** Temporary demand/supply mismatches across correlated instruments create exploitable spread divergence.
- **ETF Basket Lag:** Rebalancing latency introduces short-lived pricing gaps relative to the underlying NAV.
- **Institutional Constraints:** Large funds cannot exploit micro-dislocations due to high impact costs and risk limits.

1.2 Entry Signal Logic

A trade is initiated when all four of the following gates are simultaneously satisfied:

Gate	**Condition**	**Rationale**
Z-Score Trigger	$\|Z\| > 1.5$ std devs	Statistically significant extreme
Cointegration Gate	Engle-Granger $p < 0.10$	Confirms structural relationship
ML Confidence	XGBoost score > 0.52	Filters low-quality signals
Regime Filter	VIX + FX Vol check	Stability check

2 Backtesting Engine

The engine was implemented over 4 years of historical data (2021–2024). It explicitly models transaction costs, bid-ask spreads, and slippage.

2.1 Trade-Level Performance (QQQ / QQQM)

Due to the ultra-low volatility of these pairs, standard daily Sharpe ratios are misleading. We utilize Trade-Level Sharpe to measure actual capital risk.

Metric	**Value**	**Commentary**
Total Net P&L	+1,786 USD	Profitable after all costs
Win Rate	44.3%	Positive expectancy via asymmetry
Trade-Level Sharpe	2.148	Robust risk-adjusted return
Max Drawdown	-0.95%	Highly controlled risk
Cost Drag	**52.9%**	**Primary performance constraint**

3 Live Signal Generator

The signal engine scans all monitored pairs every 60 seconds. It completed a 48-hour stability run confirming 0% downtime and reliable logging to structured JSONL formats.

3.1 Signal Summary

- **Total Scans:** 2,910 cycles
- **Actionable Signals:** 10 (High-discipline ML filtering applied)
- **Log Integrity:** All timestamps follow ISO 8601 with full edge metadata.

4 Risk Framework

The system employs a three-layer position sizing model:

1. **Fractional Kelly (25%):** Sets the base optimal allocation.
2. **Vol Scaling:** Adjusts size to maintain a fixed 15% annual portfolio volatility.
3. **Circuit Breakers:** Strategy pauses if drawdown exceeds 5% and closes all positions at 8%.

5 Honest Limitations

- **Sharpe Calculation:** Traditional daily Sharpe is inappropriate here; trade-level metrics are used for honesty.
- **ML Generalization:** XGBoost shows signs of overfitting given small training samples; used only as a secondary gate.
- **Cost Sensitivity:** Success is highly dependent on execution quality and commission regimes.

6 Conclusion

Performance is constrained by transaction costs (52.9% drag) rather than signal quality. The Trade-Level Sharpe of 2.148 demonstrates a genuine, regime-conditional edge. With improved execution efficiency, the system is suitable for live deployment.